QUANT RESEARCH / STATISTICS / CASE STUDY

ALPHA VALIDITY

CASE STUDY

07

Alpha Validity measures how often common validation methods mistake chance for skill.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A small research group tests many trading ideas on historical prices and keeps the ones that look best. The more ideas tried the more likely a lucky pattern wins. Costs and regime changes then remove the apparent edge. This case study uses synthetic markets where the truth is known.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Research lead**	Needs to know which results can be trusted before capital is discussed.
02	**Risk reviewer**	Wants selection bias and leakage found before sign off.
03	**Quant researcher**	Needs a repeatable protocol that records every idea tried.

SCOPE AND HONEST LIMITS

- ⚠ Synthetic results do not prove tradable real world alpha.
- ⚠ Survivorship bias and data licensing must be controlled on historical data.
- ⚠ Conclusions depend on the generators and are not universal.
- ⚠ No result is claimed until the experiments have been run and reviewed.

CURRENT WORKFLOW AND FAILURE POINTS

07

PROPOSED RESEARCH / NOT YET VERIFIED

HOW THE WORK RUNS TODAY

1 TODAY
Pick an idea
A researcher chooses a signal from intuition or a paper.

→

2 TODAY
Backtest it
The signal runs once on a single historical period.

→

3 TODAY
Tune parameters
Settings are adjusted until the equity curve looks strong.

↓

4 TODAY
Keep the winner
Only the best looking variant is written up.

→

5 TODAY
Add costs late
Trading costs are applied at the end if at all.

→

6 TODAY
Present the curve
A single chart is shown with no record of discarded trials.

WHERE IT BREAKS

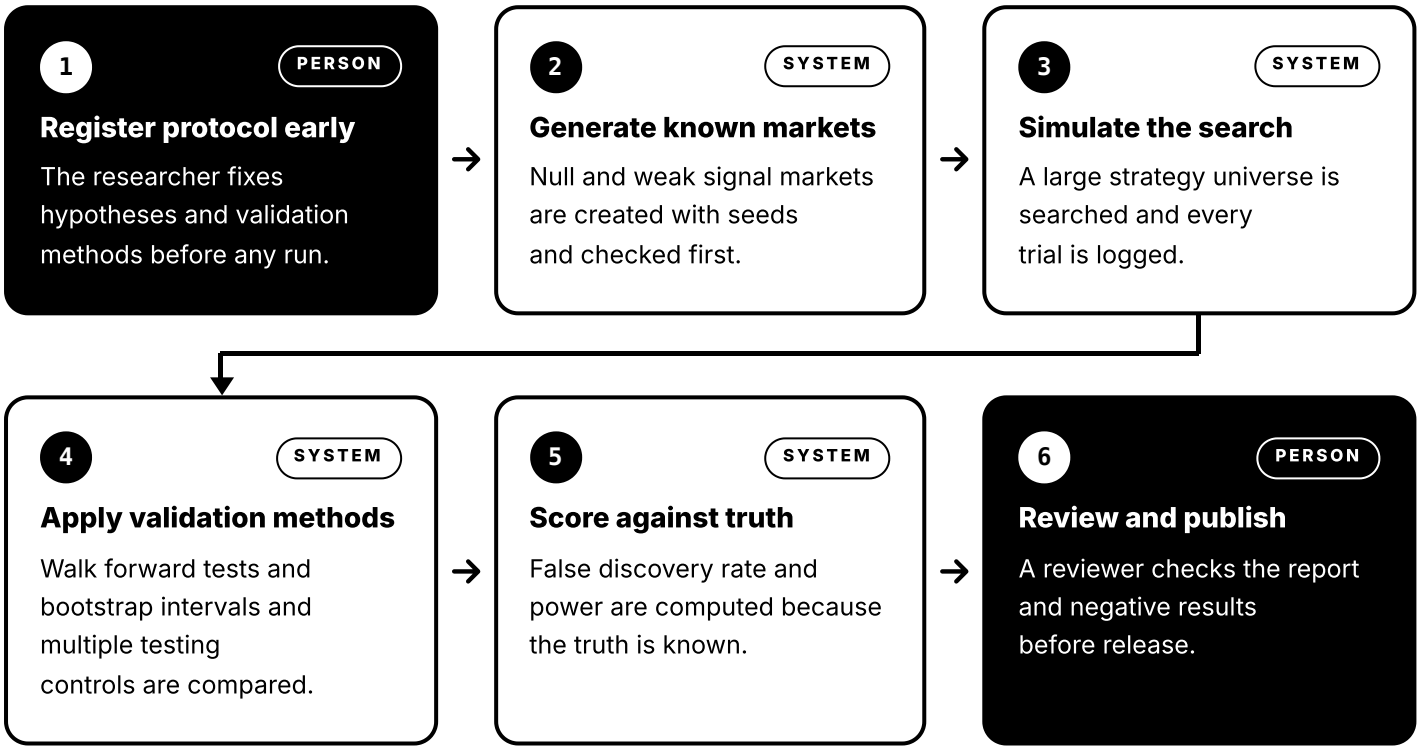
Data snooping	Many trials are run but only the best result is reported.
Overfitted parameters	Settings fit past noise and fail on new data.
Unrecorded search	The number of ideas tried is unknown so significance is overstated.
Cost blindness	Small edges vanish once realistic costs are included.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

07

PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW



SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Fix the validation protocol	RESEARCH LEAD	Choices made after seeing results would reintroduce the bias.
Judge a signal as credible	RISK REVIEWER	Statistics inform the call but a person owns the conclusion.
Release a research conclusion	RESEARCH LEAD	Publication shapes decisions so limits must be stated and approved.

ARCHITECTURE, DATA AND CONTROLS

07

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM ARCHITECTURE

1	**Synthetic markets**	Seeded generators for null and weak signal processes and regime shifts.
2	**Signal search**	Strategy universe with complete trial logging and costs.
3	**Validation lab**	Walk forward splits and bootstrap and multiple testing procedures.
4	**Scoring**	Comparison of conclusions with the known truth of each market.
5	**Statistical report**	Calibration plots and negative results and limits in a fixed template.

CORE DATA ENTITIES

- **Market scenario**
- **Strategy**
- **Trial log**
- **Validation method**
- **Test result**
- **Cost model**

RISKS AND THE CONTROL FOR EACH

Selection bias hidden by reporting →	Log every trial and apply multiple testing control to the full count.
Information leakage →	Enforce time ordered splits and test that no future data reaches a signal.
Protocol changed after results →	Freeze the protocol in the record before runs and log any later change.
Synthetic result overclaimed →	Report synthetic and licensed data findings separately with stated limits.

EVALUATION AND DELIVERY PLAN

07

PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Empirical false discovery rate	Share of accepted signals that are null in the known synthetic truth.
Statistical power	Share of true weak signals each method detects at a stated threshold.
Regime sensitivity	Change in conclusions when the market process shifts after selection.
Interval calibration	How often reported intervals contain the true effect across repeats.
Cost adjusted robustness	Whether accepted signals remain distinguishable from zero after costs.

TEST PLAN

1. **Confirm each generator reproduces its stated null or weak signal properties.**
2. **Check that time ordered splits never let future data reach a signal.**
3. **Rerun every experiment from fixed seeds and compare results exactly.**
4. **Run the full search on pure noise and confirm discoveries match chance.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree the research question and what evidence would count as a negative result.	Review how strategies are validated today and where selection enters.	Define generators and search universe and scoring and the frozen protocol.	Run the experiments and score them against truth and assemble the report.	Release the code and notebooks and the report with limits and negative findings.
DECISION Which questions are in scope	**DECISION** Which methods to compare	**DECISION** Approve the protocol	**DECISION** Accept the statistical evidence	**DECISION** Which conclusions to publish

DELIVERABLES

- **Reproducible experiment suite**
- **Trial log and protocol record**
- **Calibration plots and negative results**
- **Research notebook bundle**
- **Technical paper with stated limits**